

NIMBYism and the Housing Supply

This paper analyses an equilibrium model of housing supply. The model is populated by heterogeneous individuals who move through a life cycle. Individuals differ exogenously in their age and labor productivity. They decide endogenously about how much to consume, how much to save in liquid wealth and whether to buy a home or not. Homeownership is due to frictions in the sense that households can only finance part of their house by mortgages (loan-to-value constraint), that a house needs to have a minimum size, and that selling the house is associated with transaction costs.

Individuals value streams of consumption and housing consumption over their life cycle. When the life cycle ends, they derive additional utility from a bequest motive. If individuals do not own a home, they have to rent shelter on the competitive market. Rented homes come with a lower utility.

The baseline interesting mechanism at work in this model is a voting scheme, through which the voting population can decide whether to increase the amount of land available for building houses. The paper mainly follows a median voter idea. There is tension in the population about whether to increase housing supply: young renters and poor households generally favor additional housing supply, as this depresses prices and makes housing more affordable. Homeowners, on the other hand, want to preserve the value of their home or even increase it. Therefore, they are in favor of restricting housing supply.

The authors calibrate the model to the US economy and study how demographic changes impact on housing supply and therefore house prices. The main result is that the baby boom of the 1950s and 60s together with a currently ageing population leads to an undersupply of housing through the political process and therefore has increased house prices by around 15 percent since the 1950s.

Major comments:

Overall, I think this paper comes with an interesting idea to rationalize local housing supply. However, while the basic idea is per se interesting and may be important, the paper unfortunately falls short of one's potential. This is for three major reasons:

1. Structure of the model and assumptions

The presentation of the model lacks justification of the single model elements. The authors make a strong set of assumptions that are never explained. This includes, for example, the structure of the mortgage market, the bequest motive, the structure of the rental market, the regional structure of the model, how individuals are distributed across regions, etc. On top, the calibration of the model is only weakly discussed. In particular, I wondered to which degree the chosen moments in the “simulated moments method” can actually identify parameters.

2. Empirical credibility

It is not at all clear to me to which degree voting on local housing supply matters for house prices empirically and what I can learn from this model about it. There are several points of criticism here:

- a. The paper hardly shows any data to convince the reader of the underlying mechanism, nor compares model outcomes to empirical counterparts. As such, it remains a simple, yet interesting, theory.
- b. Voting in local elections usually has very low turnout rates (Hajnal, Z.L. (October 22, 2018). Why Does No One Vote in Local Elections? New York Times). How credible is a median voter story in light of this issue?
- c. The model has a “spatial” component, but all regions are the same. In the real world, house prices differ across regions probably as much as they differ over time. How should we understand this heterogeneity in line of the model.
- d. Housing supply is not only constrained by the political process but obviously also by land availability. How should we think about this through the lens of the model?

3. Presentation of simulation result

Actually, I felt lost when reading through the simulation results. This is because the simulation exercises are not very well explained. In particular, I wondered about the steady state results that included a demographic transition. I couldn't find out what the exact notion is here, since demographic transitions and steady states usually do not go hand in hand.